Starter Pack Analysis: C28 B2B Receivables Follow-Up Desk

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Your case ships with six invoices and one reference answer: 46000 outstanding, 36000 collectable, 10000 in review and 1 paid invoice. The fixture is thinner than the AI build spec assumes, so eight things in the spec need to change (Section 6).

1. What is in the pack

Item	What it is	Use for C28
`Starter_Data/README.txt`	Rules for all 35 fixtures	Read the rules below
`Starter_Data/C28/fixture.json`	6 invoices, one as-of date, the worked answer, 3 public checks	Reference data and tests
`Starter_Data/C28/README.txt`	Outcome, MBA opportunity, product change, unit conventions	Scope
`Starter_Data/C01` to `C35` (others)	34 other cases	Ignore
`Starter_Data/assets/P01` to `P12`	Synthetic images for image-based cases	Not used by C28
`case-bank.json`	35 case cards. The C28 card lists edge cases, pilot path and skills	Master definition
`Judge_and_Mentor_Notes.md`	How judges run and score a case	What to optimise for
`Templates/*.csv`	Rubric, case register, MBA and handoff templates	Scoring and handoff (Section 8)

Rules from the starter README

- All data is synthetic. You may add clearly labelled synthetic rows in the same schema.
- Preserve the reference inputs. Add your own negative test.
- Model calls are optional. A declared manual or mock fallback is acceptable.
- Units: `_inr` is INR, `_pct` is 0 to 100, `_units` are counts, `_days` are calendar days. Dates are ISO.
- **null means unknown, never zero.**
- Do not combine the fixture with the old MBA workbook without an explicit mapping.

2. The C28 fixture, field by field

Fixture field	Meaning	Spec name
`records.invoices[].id`	Invoice id (INV1 to INV6)	`invoice_id`
`customer`	Customer id (C1 to C6, one invoice each)	`customer`
`amount_inr`	Invoice total in INR	`amount`
`received_inr`	Total received against that invoice, one number (not a list of receipts)	applied receipts
`due`	Due date, ISO	`due_date`
`disputed`	true or false only	dispute flag
`records.as_of`	One as-of date for the whole file: 2026-10-16	default as-of
`records.worked_outstanding_inr`	6000, the reference answer for INV1	expected value, test 1
`public_checks`	Three sentences (6000 outstanding, paid leaves and disputed goes to review, as-of updates ageing)	tests 1, 2, 4, 5, 25
`synthetic`, `limit`	Labels and the "small teaching fixture, demand unproven" note	footer text

3. Reference answer at 2026-10-16

All six invoices share the due date 2026-10-01, so each is 15 days overdue (bucket 1-30). Scores use the formula in spec Section 5.7.

Invoice	Customer	Amount	Received	Outstanding	State	Score
INV1	C1	10000	4000	6000	Collectable	28.5
INV2	C2	10000	10000	0	Paid, leaves the list	none
INV3	C3	10000	0	10000	Under Review (disputed)	none
INV4	C4	10000	0	10000	Collectable	42.5
INV5	C5	10000	0	10000	Collectable	42.5
INV6	C6	10000	0	10000	Collectable	42.5

Totals: outstanding 46000, collectable 36000, under review 10000, promised 0, paid 1. Rank order: INV4, INV5, INV6 (equal scores, ordered by id), then INV1. Keep this table as a regression test (new test 42).

4. As-of boundary values (due 2026-10-01)

As-of date	Days overdue	Bucket
2026-09-30	0	Not due (all six invoices, so the priority list is empty: show an empty state, not an error)
2026-10-01	0	Not due
2026-10-02	1	1-30
2026-10-16 (fixture)	15	1-30
2026-10-31	30	1-30
2026-11-01	31	31-60
2026-11-30	60	31-60
2026-12-01	61	61-90
2026-12-30	90	61-90
2026-12-31	91	90+

5. How you will be scored

Engineering points = weight × rating (0 to 4) ÷ 4, out of 100. The MBA track is scored separately on seven dimensions.

Criterion	Weight	What it means for C28
Working journey	25	Import, set as-of, see the list, handle a dispute, review a draft, export. All end to end
Case check 1	10	10000 with 4000 received shows 6000 outstanding
Case check 2	10	Paid invoices leave the list. Disputed balances stay visible in review
Case check 3	5	Changing the as-of date updates ageing consistently

Criterion	Weight	What it means for C28
Reliability	15	Bad input handled, your own negative test, changed-input challenge passes
Business-feedback improvement	15	The negotiated change (disputes and partial receipts in priority) with before and after evidence
Handoff	10	README, tests, known gaps, backlog, contributor list
Individual technical contribution	10	Each person answers one contribution question. Keep commits per person

How judges test

- They run the reference calculation, then one changed-input challenge: change one stated input and keep the rule. The expected behaviour is written before the run.
- At the 9 AM market shock, every team gets the same window and a change taken from the card's edge cases: partial payment, unapplied receipt, disputed amount, or promise-to-pay already received. They grade the response, the trade-off, the changed product and the business consequence.
- Product negotiation and evidence can continue until submission. Only case ownership and rosters are frozen.
- A disclosed fallback is valid. Say that drafts use templates and no model.
- Judges vary inputs within the same published schema, so optional and null values must not break the app.

Likely C28 challenges and the expected behaviour

Challenge	Expected behaviour
As-of set to 2026-11-01	Every collectable invoice moves to 31-60, states unchanged
INV4 `received_inr` = 10000	INV4 becomes Paid and leaves the list
INV4 `received_inr` = 12000	Outstanding 0, flagged Overpaid by 2000
INV5 `disputed` = true	INV5 moves to Under Review, collectable falls by 10000

Challenge	Expected behaviour
`due` = null on a row	Row flagged and excluded with a message, never treated as today or zero
`received_inr` = null	Outstanding shown as unknown and flagged, never treated as 0
Same dispute or promise saved twice	No duplicate, state unchanged
`amount_inr` negative	Row rejected with a message

6. Gaps between the fixture and the AI build spec

#	Fixture reality	Spec assumed	Change
1	Field names `id`, `amount_inr`, `received_inr`, `due`, `disputed`	`invoice_id`, `amount`, `due_date` and so on	Map them in `loader.py` using the table in Section 2
2	Receipts are one number per invoice. No receipt records or dates	A list of receipts with ids and dates	Support `received_inr` and an optional `records.receipts` list. Unapplied-receipt and promise-payment logic need receipt records, so they run only on the extra synthetic file
3	`disputed` is a boolean. No amount, cause, resolution or credit	`disputed_amount`, cause, resolution, `credit_amount`	`disputed = true` with no amount holds the whole outstanding balance in Under Review with the note "dispute detail missing". Cause, resolution and credit are our optional additions, null by default, edited in the UI

#	Fixture reality	Spec assumed	Change
4	No invoice date	`credit_days`	Show credit days as unknown. Only the due date defines overdue. Accept `invoice_date` or `credit_days` if a judge adds them. Do not invent a credit term
5	One global `records.as_of`	An as-of date per invoice	One date per file. The UI picker overrides it
6	null means unknown, never zero	Not stated	A null amount, received or due flags the row and excludes it from calculations. Never default to 0
7	No promise or agreed next action in the data, although the case card lists it	Promise fields	Promise fields are optional additions, null by default
8	Extra keys (`synthetic` , `data_version` , `currency` , `limit`)	Not stated	Read `records` , ignore unknown keys, show `synthetic` and `limit` in the footer

7. Additions to make to the spec

- [ ] Add the field-mapping dictionary from Section 2 (spec Sections 4.1 and 8.1).
- [ ] Support aggregated `received_inr` and an optional receipts list (spec Sections 4.1 and 5.4).
- [ ] Add the null rule to validation (spec Section 6) with a test.
- [ ] Read the default as-of date from `records.as_of` .
- [ ] Use `worked_outstanding_inr` as the expected value in test 1.
- [ ] Add test 42: the reference run in Section 3 matches exactly. Add the boundary dates in Section 4 to the as-of tests.

- [ ] Keep `fixture.json` untouched. Put extra rows in `data/extra_synthetic.json`, labelled `"synthetic": true` and "added by team". Include a partial payment, an unapplied receipt, a disputed amount, a promise-to-pay, and your own negative test.
- [ ] Save a snapshot when the MBA team joins: a demo or screen recording, current test results, known gaps, backlog, contributor list, and the business team's joining time and baseline. Write it to `output/snapshot_at_mba_entry.md`. The case card requires this.
- [ ] State the fallback in the README and the footer: drafts use templates and no model.
- [ ] Point the handoff link in the transfer log to the repository README.

8. Templates and who uses them

Template	Purpose	Engineering link
`Judging_Rubric.csv`	The 15 criteria in Section 5 plus the MBA dimensions	Self-check before submission
`Judging.csv`	One row per judge rating	None
`Case_Register.csv`	35 cases. C28: Core level, payer is the small distributor finance lead, monthly operations subscription	Confirms the five skills in Section 9
`Venture_Brief.csv`	Customer, payer, value hypothesis, price, negotiated product change, expected test, actual result, stop rule	Supply the expected test and actual result from the evidence log
`Unit_Economics.csv` and `_Example.csv`	Base and downside per item, with a formula. The C26 example computes contribution as revenue minus costs and excludes other costs until a pilot	Supply cost-to-serve inputs, labelled as assumptions
`Growth_Plan.csv`	Weekly segment, channel, offer, budget, thresholds	MBA only

Template	Purpose	Engineering link
`Negotiation_and_Transfer_Log.csv`	Offer, counter, outcome, consent, engineering acknowledgements, prior work credited, handoff link	Acknowledge the negotiated change and give the handoff link
`Roster_Register.csv`	Participants and transfers	None

9. Case-card skills mapped to the checklist

Skill on the case card	Spec features
UI and state	F19
Data validation	F2
Tested business rules	F23
CSV or JSON handling	F1, F20
Readable evidence and a saved result	F20, F21, F22

10. Limits of the pack

- No receipt dates, promises, dispute detail or customer payment history, so a payment-reliability score is not possible. The formula in the spec uses only age, amount and broken promise.
- All six invoices have the same amount and due date. Ranking differences come only from received amounts and disputes, and INV4 to INV6 tie.
- The judging timetable in the notes (three rooms, ten teams each, seven minutes per team) is a proposal, not confirmed. Check the final timing the organisers publish.
- The pack creates no prizes, internships or production guarantees. Market demand and willingness to pay are unproven.